DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Tec

Semester : SEMESTER - 5

Subject Code & Name: BTETPE504B - EMBEDDED SYSTEM DESIGN

Time : 3 Hours] [Total Mar

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

1 An embedded system is designed for
a) General use b) Dedicated task c) High computing d) Office work

2 A System on Chip integrates
a) CPU only b) Memory only c) Multiple blocks d) I/O only

3 Functional design of an embedded system describes the
a) Hardware layout b) Internal behavior c) Physical wiring d) Final code

4 The Current Program Status Register stores the
a) Data values b) Program state c) Instruction code d) Memory address

5 The ARM pipeline improves overall
a) Memory size b) Power loss c) Execution speed d) Code length

6 Thumb instructions in ARM architecture are
a) 32-bit b) 64-bit c) Variable d) 16-bit

7 Software interrupt instructions are mainly used to access
a) Debug modes b) Hardware faults c) OS services d) Cache memory

8 The boot-loader in an embedded Linux system is responsible for
a) Running applications b) Loading kernel
c) Managing files d) Scheduling tasks

9 Kernel initialization in embedded Linux starts after the
a) Power shutdown b) User login c) Application load d) Boot-loader

10 Software timers in RTOS are managed by the
a) Compiler b) BIOS c) Kernel d) Hardware clock

11 Multitasking in embedded systems allows
a) Single execution b) Parallel tasks c) Idle operation d) Polling

12 Asynchronous communication does not require a
a) Start bit b) Stop bit c) Clock signal d) Data bit

Q2. Solve the following.

A) Discuss important design metrics of embedded systems such as performance, power, cost, and reliability.

B) Describe the embedded system design life-cycle models and explain their importance. 6

Q3. Solve the following.

A) Describe the Current Program Status Register (CPSR) and explain its significance. 6

B) Analyze the code answer the questions mentioned below: 6

```
MOVS R0, #0
MOV  R1, #5
BEQ  LABEL
MOV  R2, #1
B    END
LABEL: MOV R2, #0
END: B END
```

a) Is the branch taken by BEQ instruction?

b) Find the final value of R2.

c) Identify which instruction modified the flags.

Q4. Solve Any Two of the following.

A) Explain single-register and multiple-register load-store instructions with suitable examples. 6

B) Explain the architecture of Embedded Linux with suitable block diagram description. 6

C) Differentiate between BIOS and boot-loader in embedded Linux systems. 6

Q5. Solve Any Two of the following.

A) Describe file systems used in embedded Linux and their features. 6

B) Describe the I2C protocol, its operation, and advantages. 6

C) Describe serial communication basics and differentiate between synchronous and asynchronous communication. 6

Q6. Solve Any Two of the following.

A) Explain task management in RTOS with suitable examples. 6

B) Describe memory management techniques used in RTOS. 6

C) Discuss task notification mechanisms and their advantages over queues. 6

*** End ***